**Lahore University of Management Sciences**

Syed Babar Ali School of Science and Engineering

**AI651 — Deep Learning for Space, Time and Graphs**

# Assignment 2 Report

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**GitHub Repository Link:**

`https://github.com/FaaizKhan1999/dl4stg-assignments`

# Question 1 — River Forecasting Proposal

A directed river-graph forecaster: each station attends to its upstream sources at learned, state-dependent lags, adds announced releases, and outputs quantiles at S1–S4 for $t_0{+}10, \ldots, t_0{+}60$ min.

## 1. Architecture diagram

$B$ = issuance times, $H{=}72$ history steps, $K{=}6$ horizon steps, $N{=}7$ nodes $\{Q, \text{S1..S4}, \text{T1}, \text{TU}\}$ ($Q$ = reservoir, TU = virtual node for the unmonitored tributary), $d{=}64$, $\Delta{=}10$ min.

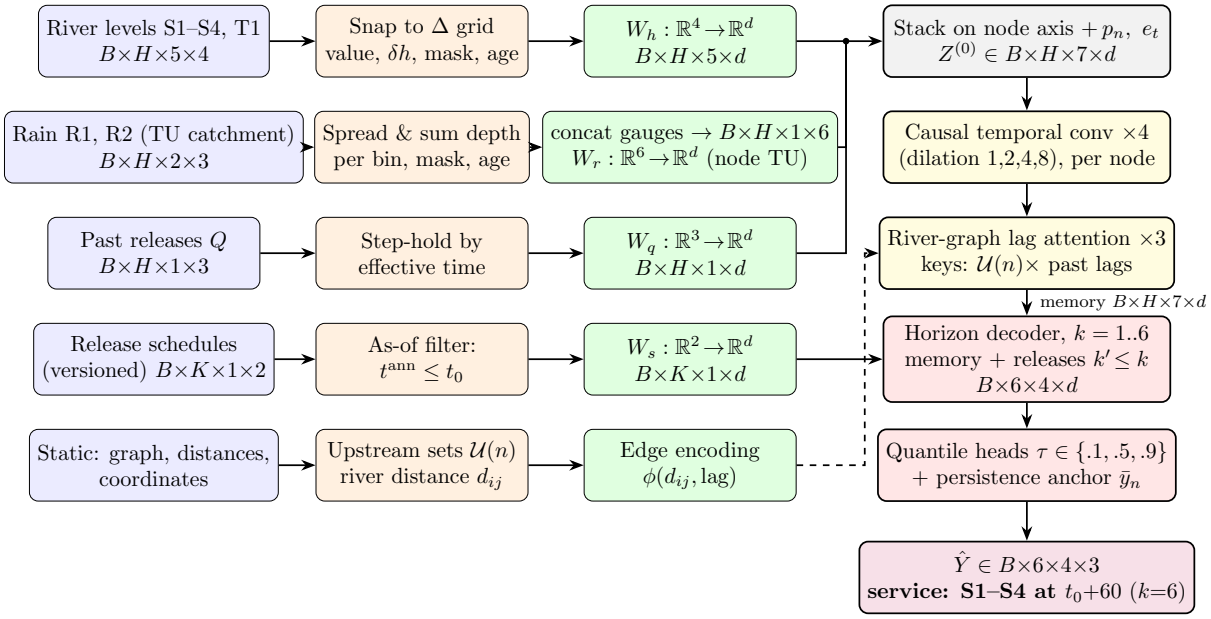



Figure 1: Pipeline. Future schedules skip the encoder and enter the decoder on the horizon axis.

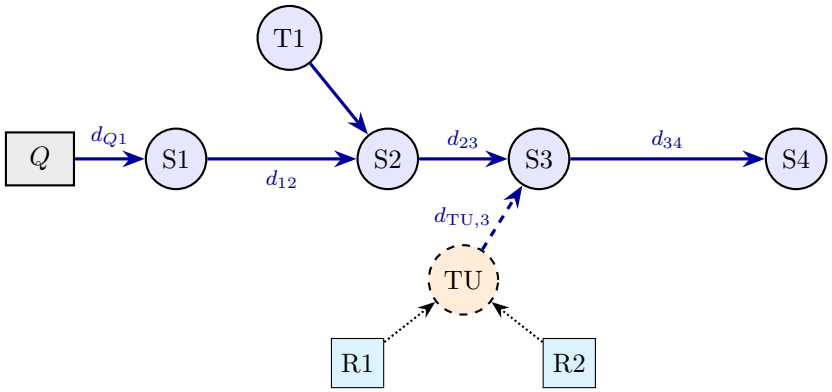



Figure 2: River graph. Solid: river edges with distances. Dotted: rain gauges feed only the virtual node TU (A7). $\mathcal{U}(n) = n$ plus every node with a path into $n$.

## 2. Assumptions

**Supplied facts:** 10-min levels at S1–S4 and T1; timestamped rain at R1, R2; recorded releases and schedules with announcement and effective times; topology and river distances. Travel times and unmonitored inflow are not supplied; future rain and levels are unavailable at issuance.

**A1** $\geq$3 years of data including several floods, so high-flow behaviour can be learned.

**A2** Levels arrive within ∼20 min and rain within ∼30 min, with arrival times logged, so inputs can be built from what had actually arrived.

**A3** Gaps are short and occasional, and faulty readings are detectable by range and rate checks, so masking is enough.

**A4** Each rain report gives the depth since the previous report, so rain is summed per bin rather than interpolated.

**A5** Everything relevant to a 1-h forecast occurred within the last 12 h, which sets $H$=72.

**A6** Flow is downstream only (no backwater), so stations attend only upstream.

**A7** R1 and R2 flank the unmonitored tributary's catchment (map), so rain enters the model only through TU and affects only S3 and S4. Rain elsewhere is either buffered by the reservoir (reaching S1 via releases) or already visible in T1's levels.

**A8** Unmonitored inflow is mainly rain-driven, so R1 and R2 are usable proxies for it.

**A9** Travel time varies smoothly with flow and distance on a stable channel, so learned lags generalise.

**A10** Release records are accurate and schedule versions are archived, which is needed for leak-free training.

**A11** Each station has a fixed level datum, so the model can predict change from the last level.

**A12** Future rain is unknown, so it is represented as uncertainty rather than as an input.

## 3. Forecasting task and data specification

$$\mathcal{I}(t_0) = \{r : t_r^{\text{obs}} \le t_0,\ t_r^{\text{arr}} \le t_0\} \cup \{\text{schedules with } t^{\text{ann}} \le t_0\},$$

$$\text{history } j = -71..0\ (12\,\text{h}), \qquad \text{horizon } k = 1..6\ \ (k{=}6:\ t_0{+}60\ \text{min}).$$

Forecasts are issued every 10 min at $t_0$ using only $\mathcal{I}(t_0)$. The outputs are the quantiles $\tau \in \{0.1, 0.5, 0.9\}$ of the level at S1–S4.

| Tensor | Dims | Axes / features | Coverage |
|---|---|---|---|
| $X^h$ levels | $B{\times}H{\times}5{\times}4$ | batch, time, node (S1–S4, T1); [level, $\delta h$, mask, age] | $(t_0{-}12\text{h}, t_0]$ |
| $X^r$ rain | $B{\times}H{\times}2{\times}3$ | batch, time, gauge; [log(1+depth), mask, age]; reshaped to $B{\times}H{\times}1{\times}6$ as TU's features | $(t_0{-}12\text{h}, t_0]$ |
| $X^q$ past releases | $B{\times}H{\times}1{\times}3$ | batch, time, $Q$; [actual, scheduled as-of, mask] | $(t_0{-}12\text{h}, t_0]$ |
| $X^s$ schedule | $B{\times}K{\times}1{\times}2$ | batch, horizon, $Q$; [release, available] | $(t_0, t_0{+}1\text{h}]$, $t^{\text{ann}}{\le}t_0$ |
| $P$; $D, M$ | $N{\times}F_p$; $N{\times}N$ | node static features; distances, upstream mask | static |
| $Y$, $m$ | $B{\times}K{\times}4$ | batch, horizon, station; target, observed-mask | $t_0{+}k\Delta$ |
| $\hat{Y}$ | $B{\times}K{\times}4{\times}3$ | batch, horizon, station, quantile | issued at $t_0$ |

**Alignment.**

- *Time:* levels take the last value in each 10-min bin; rain depth is spread over its report interval and summed per bin; releases are step-held. Masks and ages mark gaps.

- *Space:* sources are not interpolated. Each source is a node, except that the two gauges are concatenated into TU. Nodes relate only through $\mathcal{U}(n)$ and $d_{ij}$.

- *Features:* per-type projections $W_c$ map every source to width $d$ before stacking on the node axis.

## 4. Model computations and design justification

Notation: $\mathcal{U}(n)$ = upstream closure (Fig. 2), e.g. $\mathcal{U}(\mathrm{S3}) = \{\mathrm{S3}, \mathrm{S2}, \mathrm{S1}, \mathrm{T1}, Q, \mathrm{TU}\}$ and $\mathcal{U}(\mathrm{S2}) = \{\mathrm{S2}, \mathrm{S1}, \mathrm{T1}, Q\}$; $h_{t,n} \in \mathbb{R}^d$ is the representation of node $n$ at time $t$.

One this to note is that TU has no measurements, so $h_{t,\mathrm{TU}}$ is a learned rain feature, not an inflow estimate. It could be tied to actual inflow only through a constraint (e.g. on S3's rise relative to S2) or a future tributary gauge; with a gauge, TU becomes a supervised node with no structural change.

| **Stage** | **Formulation** |
|---|---|
| (a) Projection | $z^{(0)}_{t,n} = W_{c(n)} x_{t,n} + p_n + e_t$ |
| (b) Temporal | $h^{(\ell)}_{t,n} = h^{(\ell-1)}_{t,n} + \mathrm{MLP}\big(\sum_{j=0}^{2} W_j h^{(\ell-1)}_{t-j\delta_\ell,n}\big), \quad \delta_\ell = 1,2,4,8$ |
| (c) Lag attention | $\alpha_{t,j}(i,s) = \mathrm{softmax}_{i\in\mathcal{U}(j),\, s\le t} \frac{(W_Q h_{t,j})^\top (W_K h_{s,i} + \phi(d_{ij}, t-s))}{\sqrt{d}}$<br>$h_{t,j} \mathrel{+}= \sum_{i,s} \alpha_{t,j}(i,s)\, W_V h_{s,i}$ |
| (d) Decoder | $u_{k,n} \mathrel{+}= \mathrm{Attn}\big(u_{k,n}; \{h_{s,i} : i \in \mathcal{U}(n)\} \cup \{f_{k'} : k' \le k\}\big), \quad f_{k'} = W_s x^s_{k'} + e_{k'}$ |
| (e) Output | $\hat{y}^{(0.5)}_{k,n} = \bar{y}_n + \sigma_n a, \quad \hat{y}^{(0.9)} = \hat{y}^{(0.5)} + \sigma_n \,\mathrm{softplus}(b), \quad \hat{y}^{(0.1)} = \hat{y}^{(0.5)} - \sigma_n \,\mathrm{softplus}(c)$ |
| (f) Loss | $\mathcal{L} = \frac{\sum w_b\, m\, \rho_\tau(y - \hat{y}^{(\tau)})}{\sum w_b\, m}, \quad \rho_\tau(r) = \max(\tau r, (\tau-1)r), \quad w_b = 1 + \lambda \mathbb{1}[\mathrm{flood}]$ |

Table 1: Model components.

**Why each block:**

(b) Captures local trends and rain-accumulation shape.

(c) One layer reaches every upstream source at every lag (A5). The query and key depend on the current states, so lags shift with flow (A9). $\phi$ handles uneven spacing. $\mathcal{U}$ encodes flow direction (A6).

(d) Lets announced releases act on the forecast, but only on steps where they are already effective ($k' \le k$).

(e) Anchoring on the last level $\bar{y}_n$ means the model predicts change from persistence (A11). Softplus increments keep the quantiles from crossing.

(f) A point loss returns only the mean (MSE) or median (MAE). The irreducible uncertainty from the tributary and future rain needs bands, and $w_b$ up-weights floods.

Training uses chronological splits, as-of inputs, random masking, a 5-seed ensemble and conformal calibration.

| Source | Path to the forecast |
|---|---|
| Upstream stations | Keys in downstream lag attention (c), read at state-dependent lags |
| T1 | In $\mathcal{U}$(S2, S3, S4); contribution learned |
| Rainfall | R1, R2 → TU (a, b); TU is a key for S3 and S4 only (c), at a learned catchment-plus-travel lag (A7, A8) |
| Releases | Past: via $Q$ in (c). Announced future: decoder (d), mainly affecting S1 |
| **Imposed (physical)** | Topology and direction; causality ($s \le t$, $k' \le k$, no future rain); distances; rain acts only via TU; rain as depth; level continuity |
| **Learned** | Travel times, attenuation, confluence mixing, rain → TU → S3 response, release response, band widths |

## 5. Operation and exceptional conditions

| Condition | Handling |
|---|---|
| (i) Missing | Value → 0 with mask = 1 and growing age; $\bar{y}_n$ uses the last valid level. Training randomly drops points, blocks and whole stations. Missing targets are masked out of the loss. If all river data fail, fall back to persistence and alert. |
| (ii) Delayed | Only records with $t^{\text{arr}} \le t_0$ are used; a late record counts as missing until it arrives and is used in the next 10-min cycle. Training replays history as-of each issuance with logged delays (A2). |
| (iii) Announced releases | Latest schedule with $t^{\text{ann}} \le t_0$, step-held onto $k = 1..6$. Training uses archived versions, never actual future releases (that would be leakage). |
| (iv) Scheduled ≠ actual | The past input holds both actual and as-of scheduled releases, so the model learns how far to trust schedules. A live deviation appears through $Q$ within one cycle. Optional ±20% release scenarios. |

**Feasibility.** With $L = HN = 504$ tokens, even dense attention costs $2L^2d \approx 3 \times 10^7$ MACs per layer, about $10^9$ for the full 5-member ensemble. That is under 1 s on a CPU, so ingestion dominates the 10-min cycle.

## 6. Uncertainty, limitations and additional information

Uncertainty is reported as calibrated 10–90% bands (quantiles, ensemble spread and conformal calibration).

| Limitation | Impact | Response / unresolved |
|---|---|---|
| Unmonitored tributary | Unexplained rises at S3, S4 | TU rain proxy, wider bands; storms missing both gauges unresolved |
| Changing travel times | Timing errors on rising limb | State-dependent lags; floods beyond record unresolved |
| No future rain | Fast rises under-forecast | Bands widen after rain; needs nowcasts |
| Rain elsewhere unmeasured (local runoff S2–S4, T1 and S1 catchments) | Sudden local rises seen only once they reach a station | Level histories, wider bands; largely unresolved |
| Few extremes, drift | Peaks under-estimated | Event weighting, monitoring, retraining |
| Schedule deviations | Errors at S1, S2 | Deviation input, 10-min refresh |
| Sensor faults | Corrupted inputs and targets | QC and masks; slow drift partly unresolved |

| Request | Benefit | Usable without? |
|---|---|---|
| Gauge on unmonitored tributary | Supervises TU; removes the largest S3/S4 error | Yes (rain proxy) |
| Rain gauges on T1 catchment and main-river reach | Earlier warning of local runoff | Yes |
| Rain radar / nowcasts | Future-rain input | Yes, wider bands |
| Travel-time studies | Lag prior for $\phi$; validation | Yes |
| Rating curves, flood thresholds | Discharge checks, warning levels | Yes |
| Telemetry delay and QC logs | Realistic delay simulation | Yes (A2, A3) |
| Archived schedule versions | Leak-free training | Partly (lower skill) |
| Reservoir operating rules | Anticipate unscheduled releases | Yes |

## Generative AI Usage

*Generative AI tools were utilized during the completion of this assignment strictly as supplementary assistants. Primarily, AI was used as a learning aid to deepen my understanding of the assignment's underlying concepts, as well as to break down the specific architectures and techniques required for the tasks.*

*Furthermore, these tools provided support in the drafting and grammatical refinement of this report, debugging and structuring programming code, and compiling supporting evidence. However, all core ideas, critical analyses, experimental methodologies, and final implementations presented in this document are entirely my own original work. The AI was leveraged to facilitate execution and conceptual clarity, not to replace independent critical thinking or problem-solving.*